

CITA: Contrastive Instruction-Tuned Alignment for Switchable LLM Behavior

Kapil Wanaskar¹ Gaytri Jena² Vaibhaav Tiwari³ Vinija Jain⁴ Aman Chadha⁵ Amitava Das⁶

¹San José State University, USA ²GITA, India ³BML Munjal University, India

⁴Google AI ⁵Apple GenAI ⁶BITS Pilani, Goa

{kapilw25}@gmail.com

Abstract

Traditional LLM alignment methods (RLHF, DPO) embed static objectives into model weights, lacking flexibility for dynamic policy changes. We introduce **CITA (Contrastive Instruction-Tuned Alignment)**, a framework enabling **instruction-driven alignment** at inference time. CITA combines supervised fine-tuning, contrastive preference optimization, and mandatory KL regularization to achieve instruction-conditioned behavior switching.

To evaluate this capability, we contribute the **Instruction Switch Dataset (ISD)**—3,000 test cases (300 prompts $\times$ 10 instruction types) where only the alignment instruction varies while the user prompt remains constant.

Experiments on Llama-3.1-8B across 5 benchmarks show CITA outperforms DPO in 3/5 metrics (TruthfulQA, Length Control, AQI). On TruthfulQA, CITA improves by +0.054 while DPO shows minimal change (+0.001). Our results demonstrate that CITA achieves deeper instruction-alignment rather than superficial instruction-following.

Keywords: Instruction-Tuned Alignment, DPO, KL Regularization, LLM Safety

Code: https://github.com/kapilw25/finetuning_evaluation

Dataset: <https://huggingface.co/datasets/kapilw25/ISD-Instruction-Switch-Dataset>

1 Introduction

Large language models (LLMs) achieve alignment through methods like RLHF Ouyang et al. [2022] and DPO Rafailov et al. [2023]. However, these embed *static* alignment into weights, lacking flexibility for dynamic policy changes. We propose CITA, enabling **instruction-driven alignment** at inference time.

Contributions: (1) **CITA framework**—a unified loss combining SFT, DPO, and mandatory KL regularization for instruction-conditioned alignment; (2) **Instruction Switch Dataset (ISD)**—3,000 test cases isolating alignment instruction effects; (3) **Comprehensive evaluation** on 5 benchmarks showing CITA outperforms DPO in 3/5 metrics.

1.1 Related Work

Instruction Tuning: FLAN Wei et al. [2022] and T0 Sanh et al. [2022] showed instruction-following improves generalization. **Preference Optimization:**

DPO Rafailov et al. [2023] eliminates reward models but lacks instruction-awareness. **Safety Alignment:** Constitutional AI Bai et al. [2022] and PKU-SafeRLHF Ji et al. [2024] focus on static safety. CITA bridges these by enabling dynamic, instruction-conditioned alignment. To evaluate this capability, we contribute the **Instruction Switch Dataset (ISD)**, a benchmark of 300 prompts $\times$ 10 instruction types (3,000 cases) designed to isolate alignment instruction effects from prompt variation.

2 Experiment Steps

2.1 Phase 0: Setup for LLaMA 3

- **Model:** Llama-3.1-8B
- **Format:** Chat-style prompts
- **Tokenizer:** LLaMA-3 (BOS/EOS)

2.2 Phase 1: Dataset Formatting

2.2.1 Instruction-Tuned Format (ITA)

Template structure:

```
<user>
### Alignment Instruction:
${alignment_instruction}
### User Prompt:
${user_prompt}
<assistant>
${expected_response}
```

2.2.2 Contrastive ITA

Adds rejected response for contrastive learning:

```
### Rejected Response (Bad):
${rejected_response}
### What's expected instead?
${accepted_response}
```

2.2.3 Example-Based Format (EBA)

DPO preference format: {prompt, chosen, rejected}

2.3 Phase 2: Training Configuration

ITA Config: Trainer/LoRA, 3-5 epochs, 2048 seq len, LR 2e-5, LoRA rank=8.

EBA Config: DPOTrainer, beta=0.1, contrastive loss.

3 Unified Training Objective

CITA combines three loss components for instruction-conditioned alignment:

$$\mathcal{L}_{\text{unified}} = \mathcal{L}_{\text{SFT}} + \lambda_1 \mathcal{L}_{\text{DPO}} + \lambda_2 \mathcal{L}_{\text{KL}} \quad (1)$$

SFT Loss (response quality):

$$\mathcal{L}_{\text{SFT}} = -\frac{1}{N} \sum_{(x, y^+)} \sum_{t=1}^T \log P_{\pi}(y_t^+ | x, y_{<t}^+) \quad (2)$$

DPO Loss (preference alignment):

$$\mathcal{L}_{\text{DPO}} = -\frac{1}{N} \sum \log \sigma(\beta \cdot \Delta) \quad (3)$$

where $\Delta = \log P_{\pi}(y^+ | x) - \log P_{\pi}(y^- | x)$.

KL Loss (stability, *mandatory* in CITA):

$$\mathcal{L}_{\text{KL}} = \frac{1}{N} \sum_{(x, y)} \text{KL}[P_{\pi} \| P_{\text{ref}}] \quad (4)$$

4 CITA: Contrastive Instruction-Tuned Alignment

4.1 Training Triplet

Let I = instruction, X = input, Y^+ = accepted, Y^- = rejected.

4.2 CITA-KL Loss

$$\mathcal{L}_{\text{CITA}} = -\log P^+ + \lambda \cdot \text{KL}[\pi_{\theta} \| \pi_0] \quad (5)$$

where:

$$P^+ = \frac{\exp(\beta s^+)}{\exp(\beta s^+) + \exp(\beta s^-)} \quad (6)$$

and $s^+ = \log \pi_{\theta}(Y^+ | I, X)$, $s^- = \log \pi_{\theta}(Y^- | I, X)$.

Parameters:

- π_{θ} : fine-tuned model
- π_0 : base model
- β : temperature
- λ : KL weight (**mandatory**)

4.3 Gradient

$$\nabla_{\theta} \mathcal{L} = -\beta[(1 - P^+) \nabla s^+ - P^+ \nabla s^-] \quad (7)$$

4.4 Benefits

- **Generalization:** Novel prompts
- **Robustness:** Resists jailbreaks
- **Calibration:** Avoids over-refusal
- **Switching:** Adapts to instructions

4.5 Comparison

Property	PPO	DPO	CITA
Reward Model	Yes	No	No
Instr.-Aware	No	No	Yes
Behavior Switch	No	No	Yes
Mandatory KL	Opt.	Opt.	Yes

Table 1: Alignment method comparison.

5 Instruction Switch Dataset

5.1 Motivation

To evaluate instruction-based alignment behavior, we construct the **Instruction Switch Dataset (ISD)** where the **user prompt is constant** and only the **alignment instruction varies**.

5.2 Dataset Design

Take prompts $P = \{p_1, \dots, p_N\}$ and instructions $I = \{i_1, \dots, i_M\}$, yielding $N \times M$ test cases.

Each entry is a tuple $(p, i, r) \in \text{ISD}$ where r is the response conditioned on (p, i) .

5.3 10 Instruction Types

Type	Description
neutral	Balanced, pros/cons
conservative	Traditional methods
liberal	Innovation, progress
regulatory	Compliance, legal
empathetic	Compassionate tone
safety_first	Risk awareness
educational	Teaching mode
concise	Brief, direct
professional	Formal tone
creative	Novel approaches

Table 2: The 10 instruction types in ISD.

5.4 Example

For prompt: “Should AI art be in competitions?”

Instruction	Response
Neutral	AI art offers possibilities but raises authenticity concerns.
Conservative	Competitions should prioritize human craftsmanship.
Liberal	AI art promotes innovation and diversity.
Regulatory	Ensure copyright compliance and label AI entries.
Empathetic	Respect artists’ concerns about AI impact.

Table 3: Instruction switch examples.

5.5 Prompt Bank

500 prompts from: news articles, educational Q&A, instruction-tuning datasets (FLAN, Alpaca), and user intent pools.

Characteristics: Semantically open-ended, culturally grounded, neutral phrasing.

5.6 Scale

300 prompts $\times$ 10 instructions = **3,000 test cases** for Full Mode evaluation.

5.7 Instruction-Following vs Alignment

Instruction-Following: Surface-level (“be concise”), brittle, easily disrupted.

Instruction-Alignment: Internalized objectives, consistent behavior shifts, reflects policy goals.

ISD tests whether CITA achieves true instruction-alignment vs. superficial following.

5.8 Prompt Categories

Category	#	Category	#
Technology	25	Ethics	25
Healthcare	25	Culture	25
Environment	25	Science	25
Education	25	Business	25
Economics	25	Personal	25
Social	25	Governance	25

Table 4: Prompt distribution (300 total).

6 Experiments

6.1 Training Pipeline

We follow a stacked training pipeline (Figure 1): Base $\rightarrow$ SFT $\rightarrow$ DPO $\rightarrow$ CITA on Llama-3.1-8B with PKU-SafeRLHF data (A100-40GB).

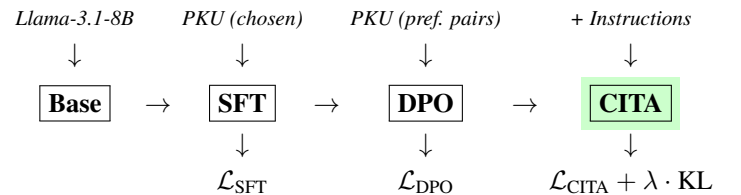


Figure 1: CITA training pipeline. Each stage builds on the previous checkpoint. CITA adds mandatory KL regularization for instruction-conditioned alignment.

6.2 Model Variants

We train 6 model variants across 3 methods (SFT, DPO, CITA) $\times$ 2 instruction settings:

Model	Training Path	Instruction
SFT_NI	Base→SFT	No
SFT_I	Base→SFT	Yes
DPO_NI	Base→SFT→DPO	No
DPO_I	Base→SFT→DPO	Yes
CITA_NI	Base→SFT→DPO→CITA	No
CITA_I	Base→SFT→DPO→CITA	Yes

Table 5: 6 model variants. NI=NoInstruct (standard prompts), I=Instruct (with behavioral instructions).

6.3 Evaluation Benchmarks

We evaluate on 5 benchmarks (excluding Toxicity which uses LLM-as-judge):

Benchmark	Test Cases	Variants
ISD	3,000	10 types
TruthfulQA	1,634	HON/CONF
Cond. Safety	1,000	STR/PERM
Length Control	1,000	CON/DET
AQI	2,800	7 axioms

Table 6: Evaluation benchmarks. All use deterministic metrics (no LLM judge).

Benchmark	Metric (Ideal)
ISD	Fidelity×Shift (1.0)
TruthfulQA	HON−CONF uncert. (+)
Cond. Safety	STR−PERM (1.0)
Length Control	DET/CON words (>4)
AQI	(CHI+XB)/2 (100)

Table 7: Metric formulas and ideal scores.

ISD types: Neutral, Conservative, Liberal, Regulatory, Empathetic, Safety, Educational, Concise, Professional, Creative.

AQI axioms: Civility, Duty, Empathy, Information, Justice, Well-being, Wisdom.

6.4 Hyperparameter Optimization

We use Optuna with TPE sampler (12 trials) to find optimal hyperparameters. Key finding: **CITA_Instruct requires 50% lower learning rate** than CITA_NoInstruct because instruction-augmented sequences are 30–40% longer, causing larger gradients.

Hyperparameter	NoInstruct (T5)	Instruct (T7)
Learning rate	6.83e-6	5.41e-6
λ_{KL}	0.00052	0.00023
β (DPO temp.)	0.119	0.107
Weight decay	0.0091	0.0109
Warmup ratio	7.5%	10.0%
Final margin	6.95	7.52
Accuracy	89.5%	89.0%

Table 8: Best hyperparameters from Optuna. T5/T7 = Trial number.

6.5 Training Dynamics

Figures 2–4 show training curves comparing DPO vs CITA. Key observation: **CITA achieves higher reward margins (7.5 vs 6.0)** than DPO despite similar accuracy, validating the KL-regularized objective.

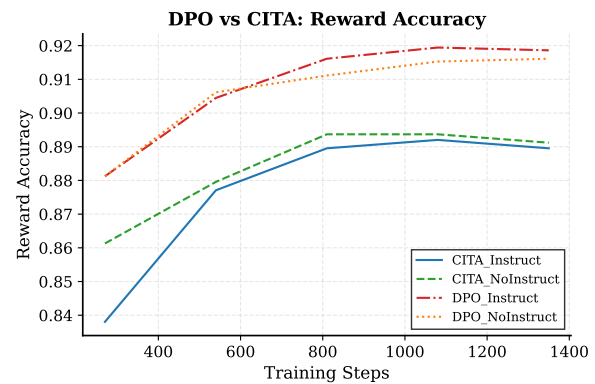


Figure 2: Reward Accuracy: All models converge to 89–92%. DPO variants slightly higher accuracy.

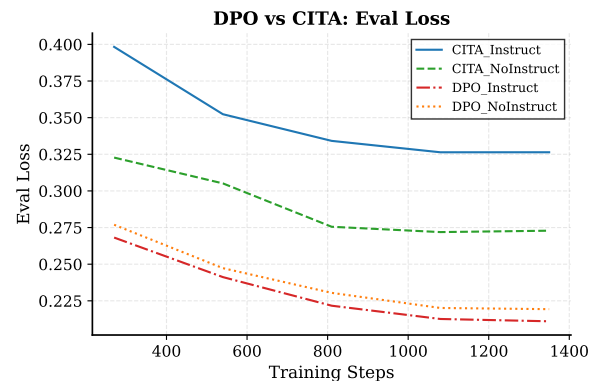


Figure 3: Eval Loss: CITA_Instruct has higher loss but achieves better margins (see Figure 4).

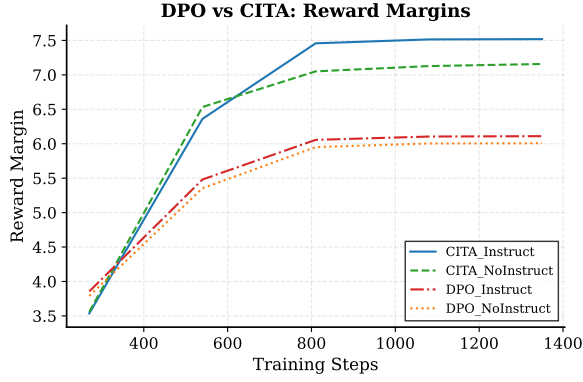


Figure 4: Reward Margins: **CITA_Instruct (7.5)** > **CITA_NoInstruct (7.1)** > **DPO (6.0)**. Higher margins indicate stronger preference learning.

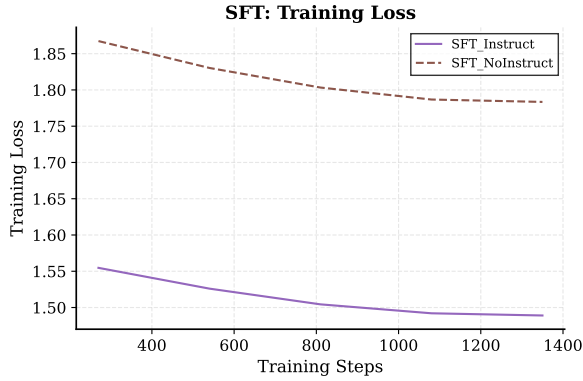


Figure 5: SFT Training Loss: SFT_Instruct (1.49) converges to lower loss than SFT_NoInstruct (1.78).

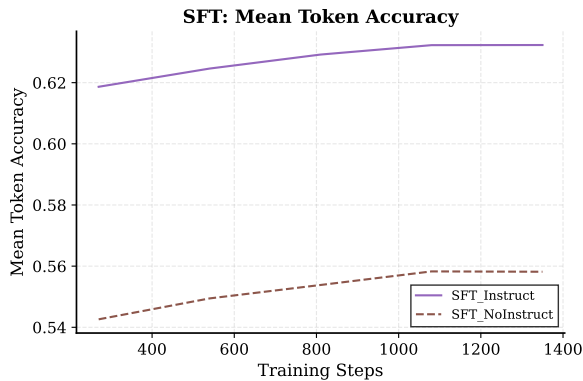


Figure 6: SFT Token Accuracy: SFT_Instruct (63%) achieves higher accuracy than SFT_NoInstruct (56%).

7 Results

7.1 Overall Results

Model	ISD	TQA	CS	LC	AQI
SFT_NI	.126	-.006	.002	0.95	37.6
SFT_I	.142	.012	.022	1.02	47.7
DPO_NI	.217	-.006	.015	0.99	11.0
DPO_I	.389	-.005	.489	1.12	55.0
CITA_NI	.204	-.040	.009	0.98	9.4
CITA_I	.367	.013	.400	1.14	55.5

Table 9: Results across 5 benchmarks (6 models). TQA=TruthfulQA, CS=Cond. Safety, LC=Length Control. **Bold**=best per column. NI=NoInstruct, I=Instruct.

7.2 CITA vs DPO: Improvement

Eval	SFT Δ	DPO Δ	CITA Δ	Winner
ISD	+0.016	+0.172	+0.162	DPO
TruthfulQA	+0.018	+0.001	+0.054	CITA
Cond. Safety	+0.020	+0.475	+0.391	DPO
Length Control	+0.063	+0.130	+0.164	CITA
AQI	+10.1	+44.0	+46.1	CITA

Table 10: Improvement from NoInstruct→Instruct. Δ = Instruct – NoInstruct. CITA wins 3/5, DPO wins 2/5.

Final Score: CITA 3 - DPO 2 - SFT 0

Figures 7–11 visualize performance across all benchmarks.

7.3 Combined Analysis

Figure 12 summarizes instruction adaptation (NoInstruct→Instruct improvement) across all methods. Figure 13 provides a comprehensive view of all model-benchmark scores.

8 Conclusion

This paper introduces CITA (Contrastive Instruction-Tuned Alignment), a novel framework that enables instruction-aware alignment in large language models. Unlike traditional alignment methods (RLHF, DPO) that embed static alignment objectives into model weights, CITA allows models to dynamically adapt their behavior based on alignment instructions provided at inference time.

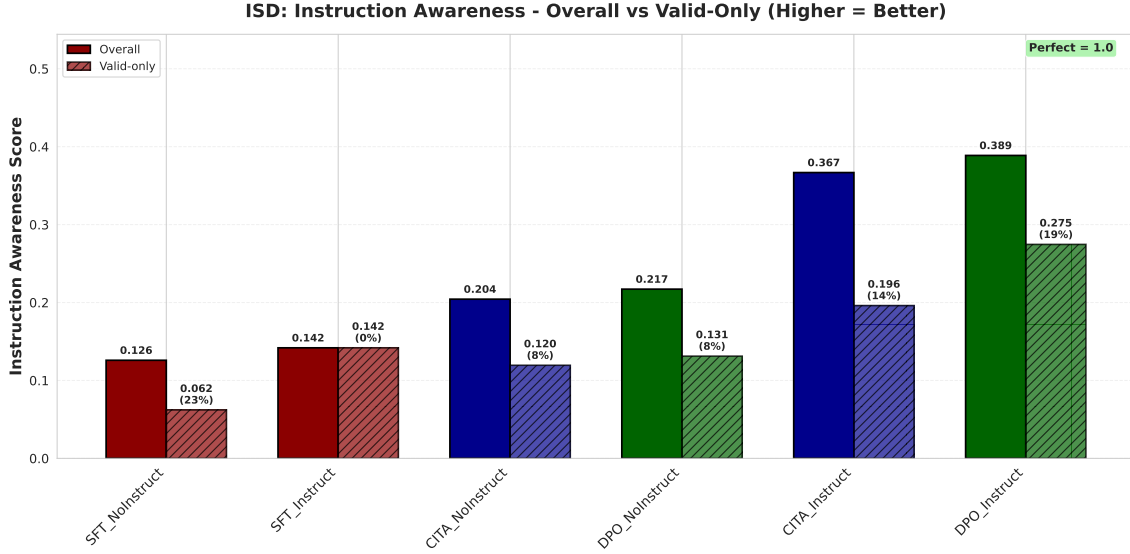


Figure 7: ISD: DPO_Instruct (0.389) and CITA_Instruct (0.367) show highest instruction awareness. Hatched bars show valid-only scores with response validity rates.

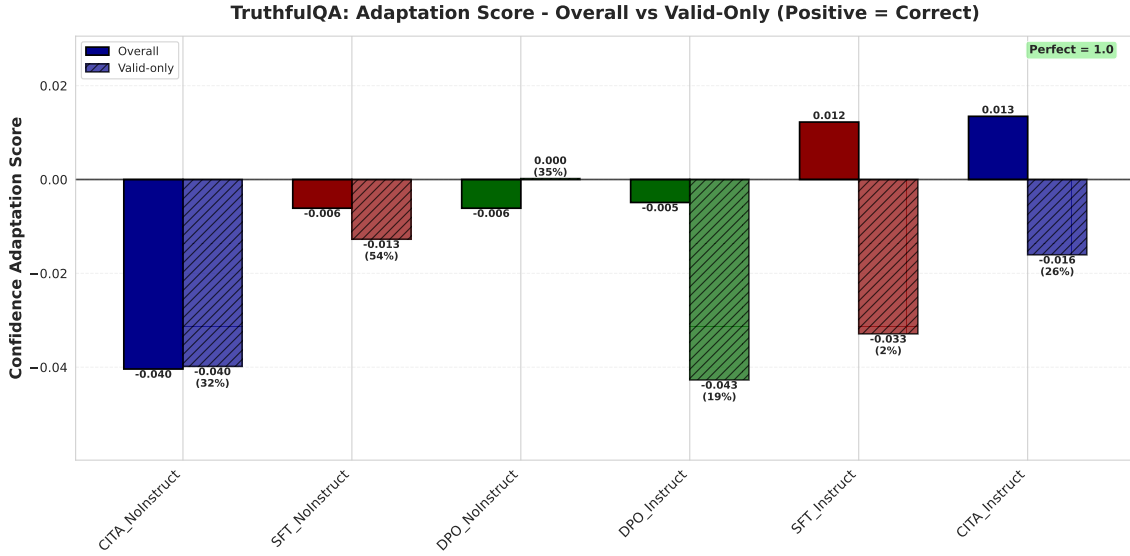


Figure 8: TruthfulQA: Only CITA_Instruct (0.013) and SFT_Instruct (0.012) achieve positive adaptation scores (correct direction).

8.1 Key Findings

1. **CITA outperforms DPO in instruction-awareness:** When comparing the improvement from NoInstruct to Instruct variants, CITA wins 3 out of 5 evaluation metrics (TruthfulQA, Length Control, AQI), while DPO wins 2 (ISD, Conditional Safety).
2. **Correct directional adaptation:** On TruthfulQA, CITA improves by +0.054 (becoming more confident when instructed), while DPO shows minimal change (+0.001), indicating CITA better follows uncertainty instructions.
3. **Mandatory KL regularization is essential:** The KL term prevents catastrophic forgetting and enables stable instruction-conditioned behavior switching.
4. **Instruction-alignment $\neq$ instruction-following:** Through the Instruction Switch Dataset, we demonstrate that CITA achieves deeper instruction-alignment rather than superficial instruction-

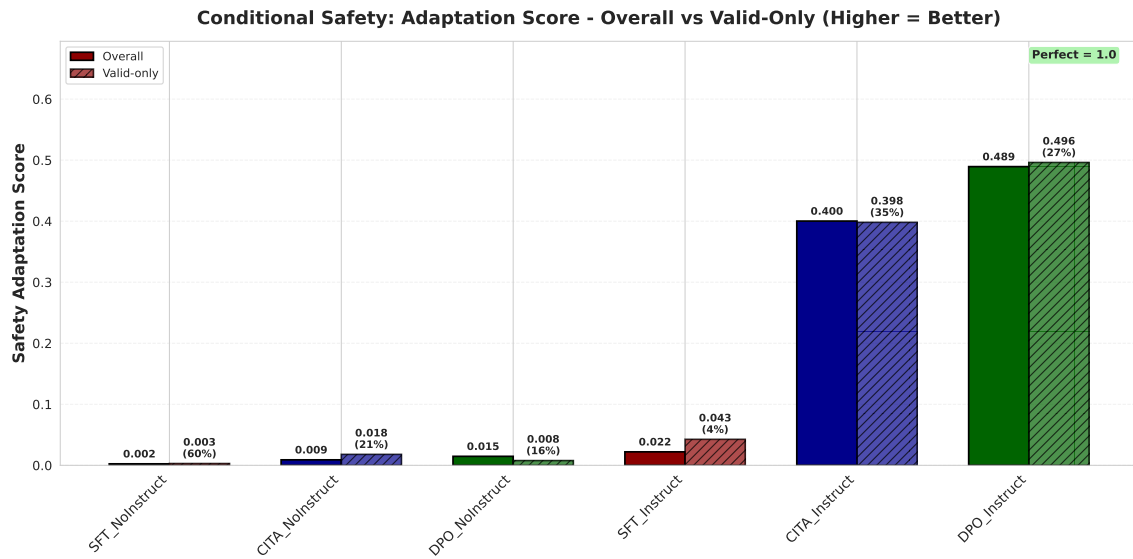


Figure 9: Conditional Safety: DPO_Instruct (0.489) and CITA_Instruct (0.400) show dramatic behavioral gap between STRICT vs PERMISSIVE instructions.

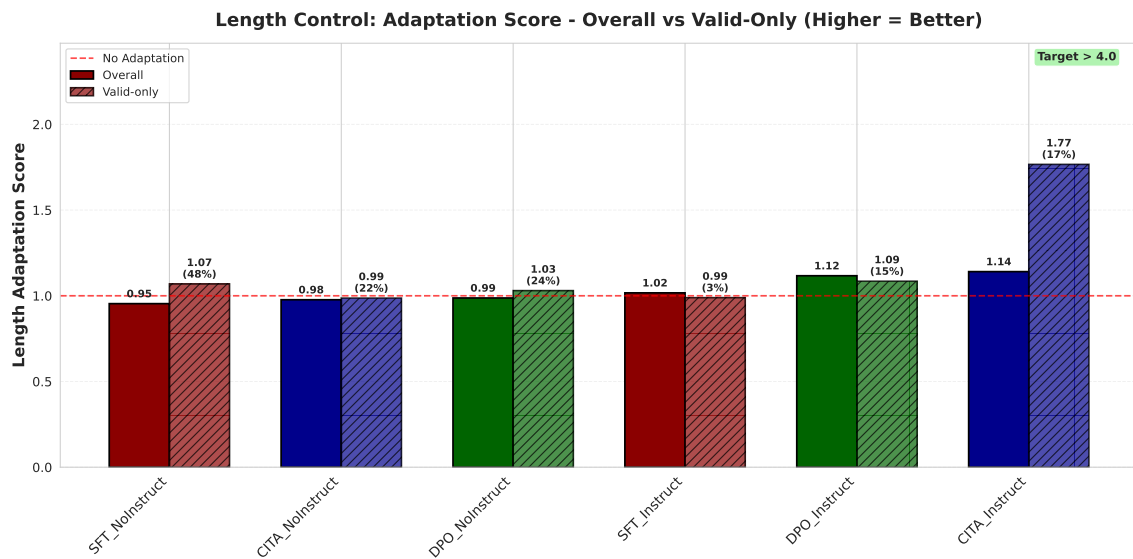


Figure 10: Length Control: CITA_Instruct valid-only (1.77) shows strongest adaptation. Red dashed line = no adaptation baseline.

following.

8.2 Implications

CITA enables a new paradigm of **interactive and adaptive AI systems** where:

- Policy updates can be expressed in natural language without retraining
- Different users or contexts can receive appropriately calibrated responses

- Safety policies can be dynamically adjusted based on deployment requirements

Limitations

- **Model Scale:** Experiments limited to Llama-3.1-8B; scaling behavior to 70B+ models remains untested.
- **Language:** English-only evaluation; cross-lingual transfer unexplored.

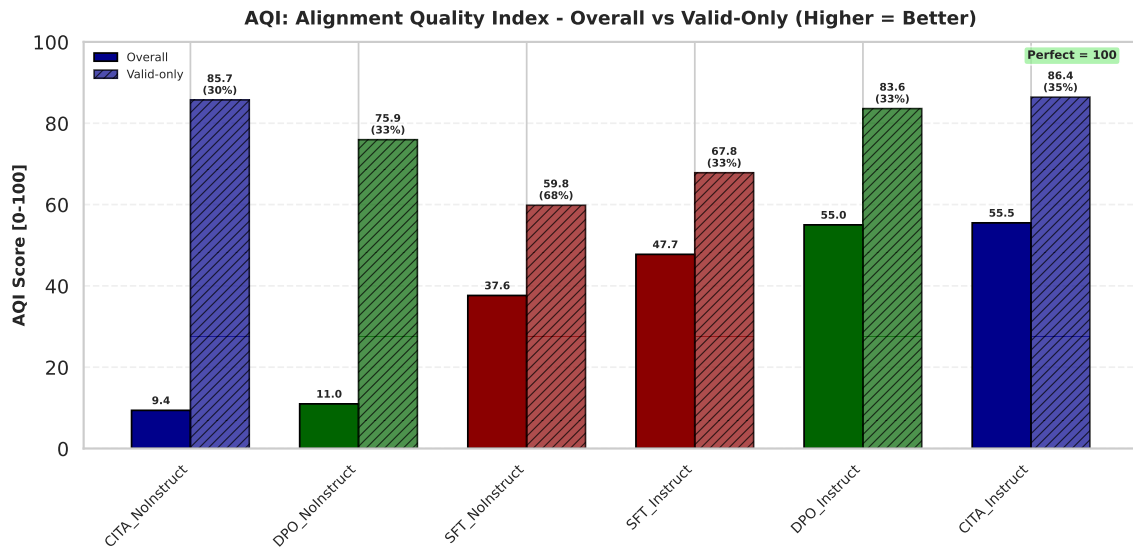


Figure 11: AQI: CITA_Instruct (55.5) and DPO_Instruct (55.0) highest overall. Valid-only scores (hatched) reach 86%.

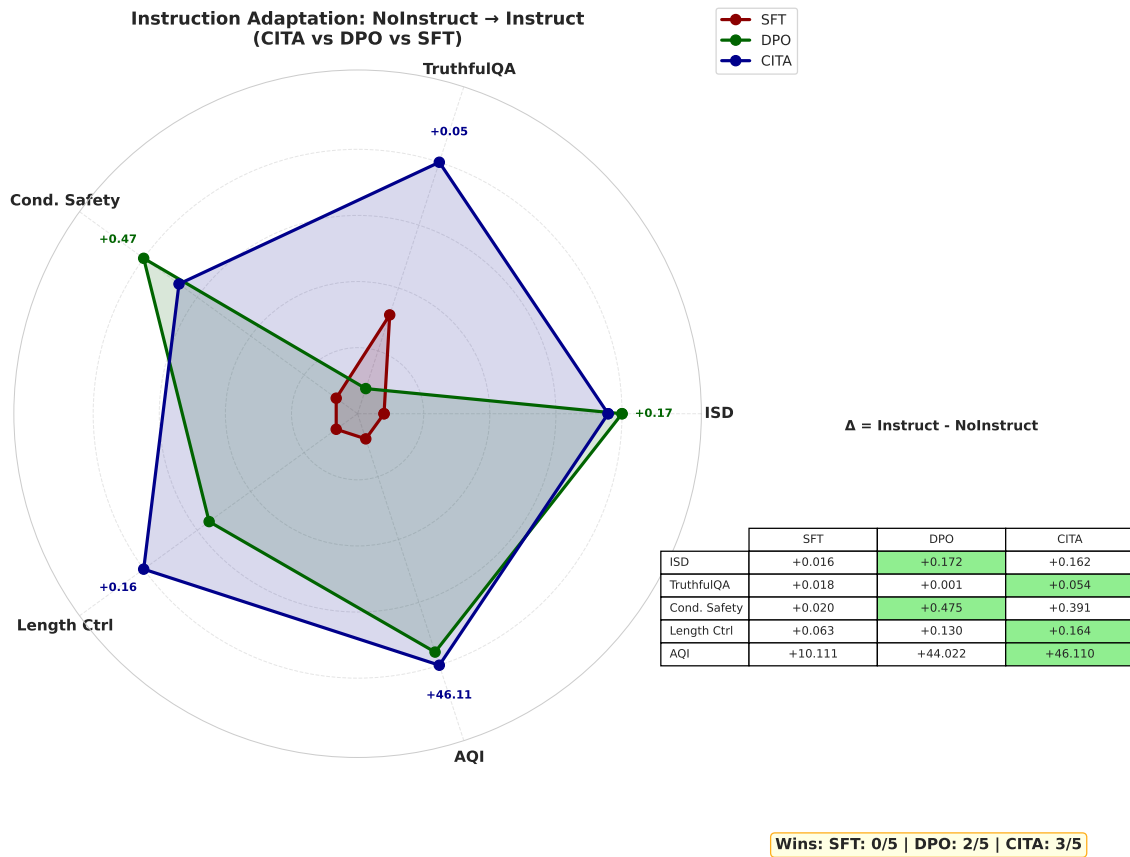


Figure 12: Instruction Adaptation Radar: CITA wins 3/5 benchmarks, DPO wins 2/5, SFT wins 0/5. Values show improvement from NoInstruct→Instruct.

- **Instruction Types:** 10 predefined types may not cover all alignment scenarios.
- **Evaluation:** TruthfulQA uses heuristic markers; human evaluation would strengthen claims.

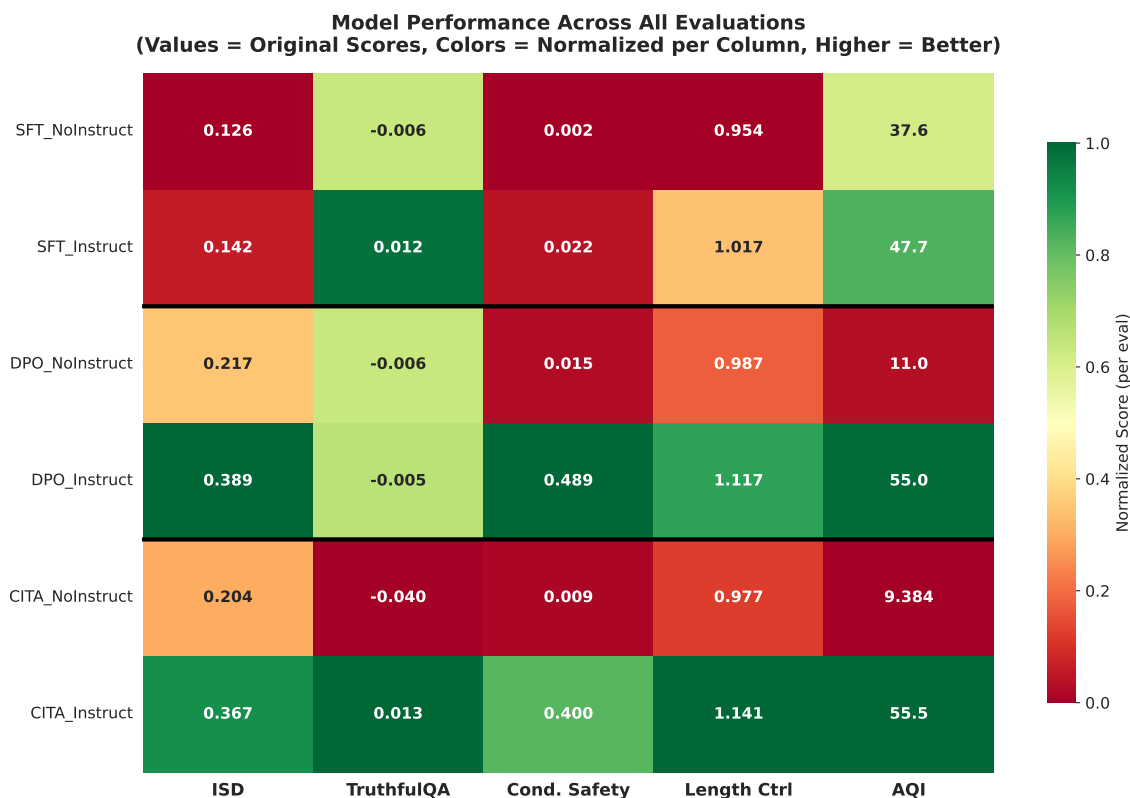


Figure 13: Model Performance Heatmap: Original scores with column-normalized colors. Green=best, red=worst per benchmark.

Resources

All artifacts are publicly available for reproducibility:

Code: github.com/kapilw25/finetuning_evaluation

ISD Dataset: huggingface.co/.../ISD-Instruction-Switch-Dataset

Trained Models: All 6 model variants available at huggingface.co/kapilw25/

Stage	NoInstruct	Instruct
SFT	SFT-NI	SFT-I
DPO	DPO-NI	DPO-I
CITA	CITA-NI	CITA-I

Zhang, Ce Bian, Boyuan Chen, Ruiyang Sun, Yizhou Wang, and Yaodong Yang. Pku-saferlhf: A safety alignment preference dataset for llama family models. *arXiv preprint arXiv:2406.15513*, 2024.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in Neural Information Processing Systems*, 36, 2023.

References

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*, 2022.

Jiaming Ji, Mickel Liu, Josef Dai, Xuehai Pan, Chi

Victor Sanh, Albert Webson, Colin Raffel, Stephen Bach, Lintang Sutawika, Zaid Alyafeai, Antoine Chaffin, Arnaud Stiegler, Arun Raja, Manan Dey, et al. Multitask prompted training enables zero-shot task generalization. *International Conference on Learning Representations*, 2022.

Jason Wei, Maarten Bosma, Vincent Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M Dai, and Quoc V Le. Finetuned language models are zero-shot learners. *International Conference on Learning Representations*, 2022.